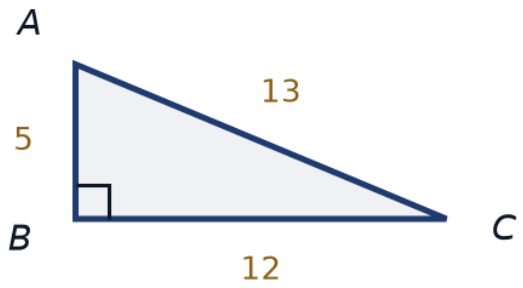
Right-Triangle Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Trigonometric ratios

A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:



a $\sin\theta$ **b** $\cos\theta$ **c** $\tan\theta$

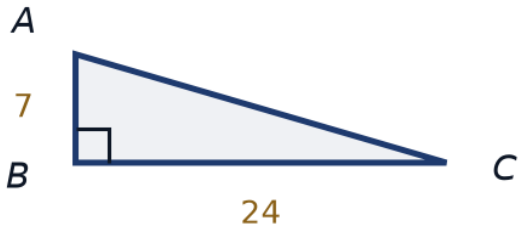
2 [Foundational] • Target: Trigonometric ratios

Write the exact value of each expression:

a $\sin 30°$ **b** $\cos 45°$ **c** $\tan 60°$ **d** $\sin 60°$

3 [Foundational] • Target: Trigonometric ratios

A right triangle has legs of length 7 and 24.



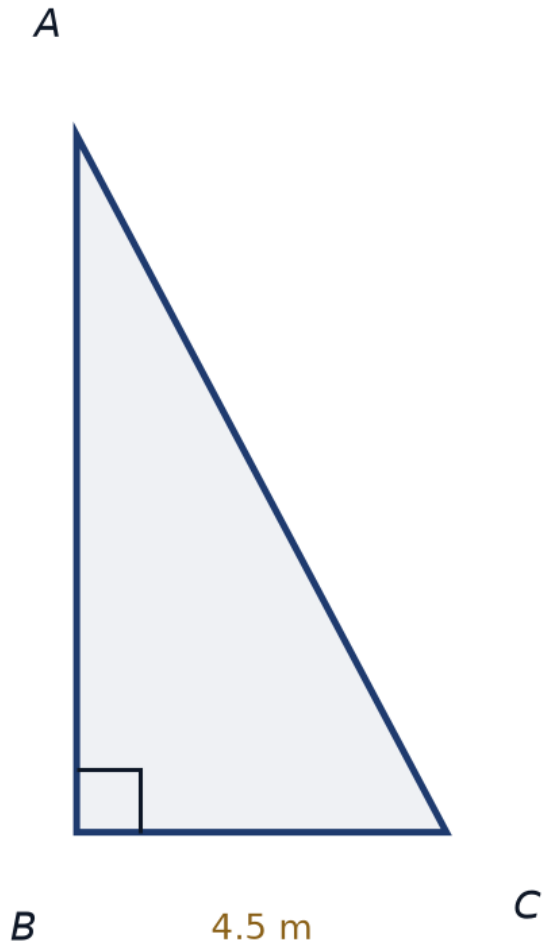
a Find the length of the hypotenuse.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.

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4 [Foundational] • Target: Applications

A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.



5 [Foundational] • Target: Applications

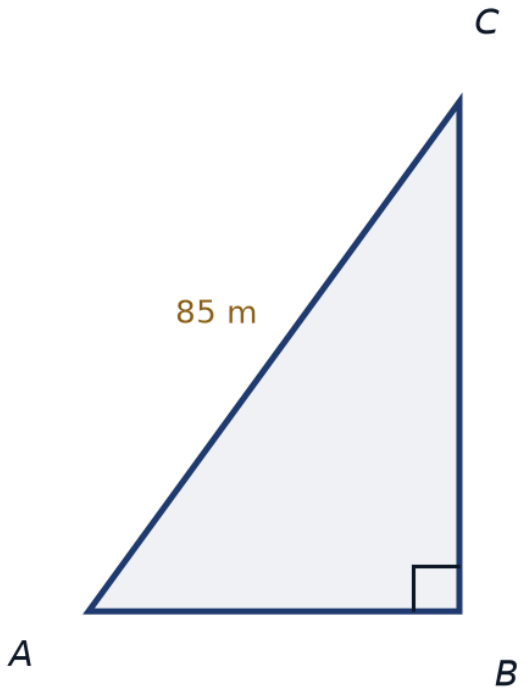
From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.



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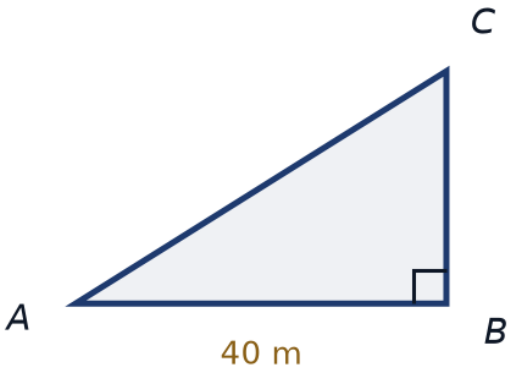
6 [Foundational] • Target: Applications

A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.



7 [Foundational] • Target: Applications

Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is $32°$. Find the height of the taller building, to one decimal place.

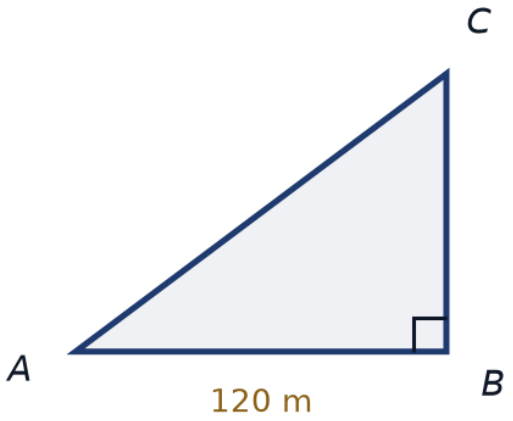


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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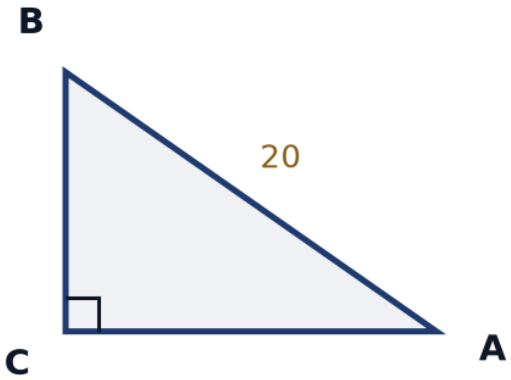
8 [Intermediate] • Target: Applications

A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as $37°$. Find the height of the tower, to the nearest tenth of a metre.



9 [Intermediate] • Target: Solving right triangles

In right triangle ABC with the right angle at C, $A = 35°$ and the hypotenuse $AB = 20$. Find:



a side BC (opposite A)

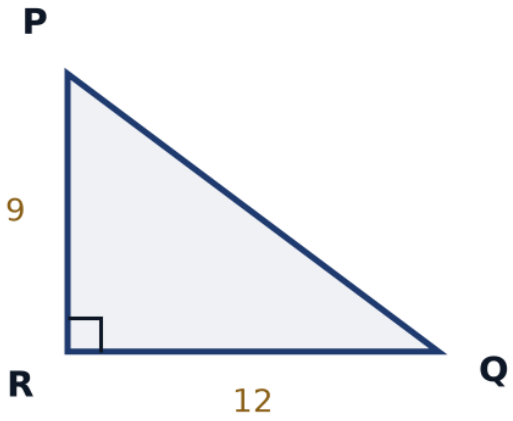
b side AC (adjacent to A)

c angle B

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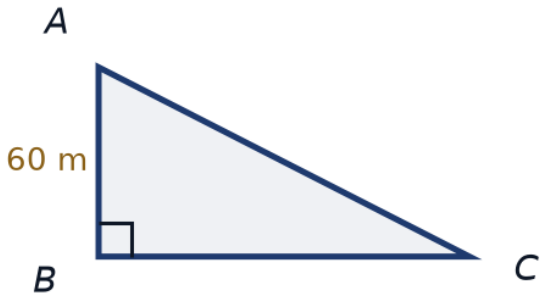
10 [Intermediate] • Target: Solving right triangles

In right triangle PQR with the right angle at R, the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.



11 [Intermediate] • Target: Angles of elevation and depression

A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is $27°$. Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.



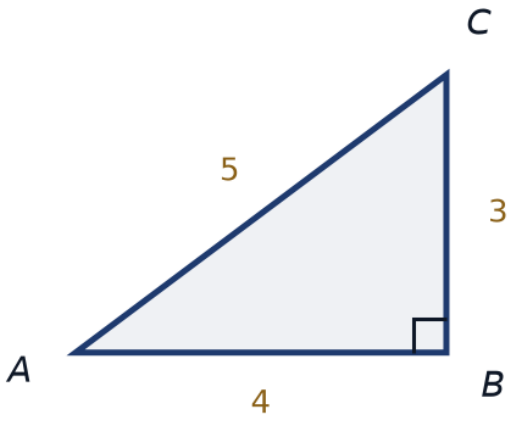
12 [Intermediate] • Target: Angles of elevation and depression

From a lighthouse 45 m tall, the angle of depression to a ship is $12°$, and a second ship is spotted further out with angle of depression $7°$. Find the distance between the two ships, to the nearest metre.

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13 [Intermediate] • Target: The reciprocal trig ratios

For a right triangle with opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$ (relative to angle θ), find:



a $\csc\theta$ **b** $\sec\theta$ **c** $\cot\theta$

14 [Intermediate] • Target: The reciprocal trig ratios

Explain why $\csc\theta = \frac{1}{\sin\theta}$, and use this to find $\csc(30°)$.

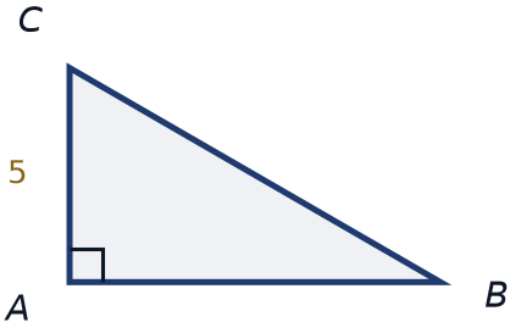
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 6.02 Polar Equations and Graphs

A surveyor at point A measures the angle of elevation to a mountain peak as $22°$. Walking 500 m directly toward the mountain to point B, the angle of elevation increases to $35°$. Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)

16 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

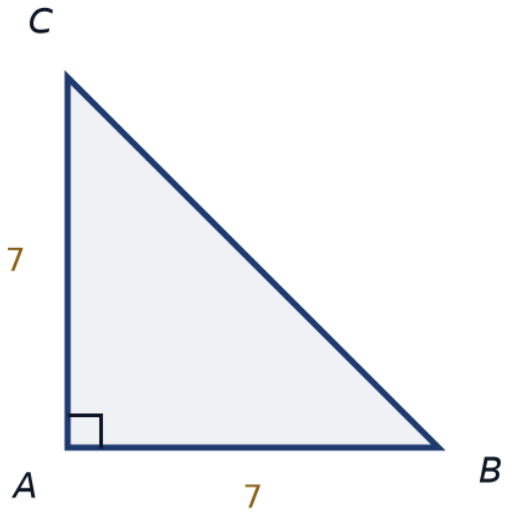
In a 30-60-90 triangle, the shortest side (opposite the $30°$ angle) has length 5. Find the lengths of the other two sides in exact form.



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17 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.02 Logarithms and Their Properties

In a $45\text{-}45\text{-}90$ triangle, each leg has length 7. Find the hypotenuse in exact form.

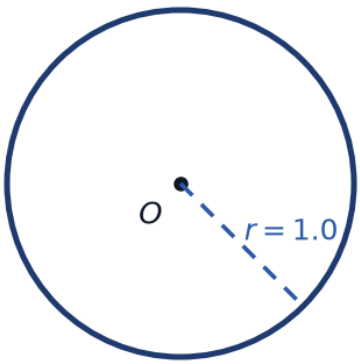


18 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A right triangle used to define $\sin\theta$ and $\cos\theta$ only works directly for acute angles θ. Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.

19 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.04 Applications: Growth and Decay

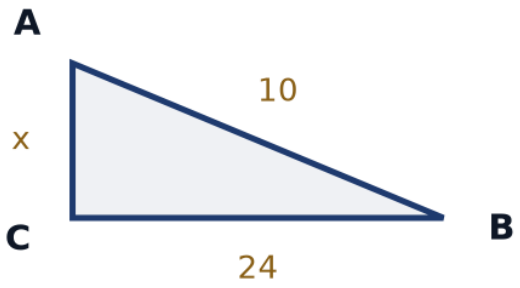
Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



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20 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 8.01 Rational Functions and Asymptotes

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Six Trigonometric Ratios

In a right triangle with acute angle θ, the side opposite θ is 5 and the hypotenuse is 13. Find $\cos\theta$.

a. 12/13

b. 5/12

c. 13/12

d. 12/5

22 [Foundational] • Target: The Six Trigonometric Ratios

For the same triangle ($extopp = 5$, $\text{adj} = 12$, $\text{hyp} = 13$), find $\sec\theta$.

a. 13/12

b. 13/5

c. 12/13

d. 12/5

23 [Foundational] • Target: The Six Trigonometric Ratios

If $\tan\theta = 3/4$ in a right triangle, find $\csc\theta$.

a. 5/3

b. 5/4

c. 4/3

d. 3/5

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24 [Foundational] • Target: Special Right Triangles (30-60-90 and 45-45-90)

In a $30^\circ - 60^\circ - 90^\circ$ triangle, if the shorter leg is 4, find the lengths of the longer leg and the hypotenuse.

a. Longer leg $= 4\sqrt{3}$, Hypotenuse $= 8$

b. Longer leg $= 8$, Hypotenuse $= 4\sqrt{3}$

c. Longer leg $= 4\sqrt{2}$, Hypotenuse $= 8$

d. Longer leg $= 6$, Hypotenuse $= 8$

25 [Foundational] • Target: Special Right Triangles (30-60-90 and 45-45-90)

In a $45^\circ - 45^\circ - 90^\circ$ triangle, if the hypotenuse is 10, find the length of each leg.

a. $5\sqrt{2}$

b. 5

c. $10\sqrt{2}$

d. $5\sqrt{3}$

26 [Foundational] • Target: Special Right Triangles (30-60-90 and 45-45-90)

What is the exact value of $\sec(60^\circ) + \tan(45^\circ)$?

a. 3

b. $2 + \sqrt{3}$

c. $1 + \sqrt{3}$

d. 2

27 [Foundational] • Target: Cofunction Identities and Reciprocals

Which cofunction identity correctly relates sine and cosine for complementary acute angles?

a. $\sin(90^\circ - \theta) = \cos\theta$

b. $\sin(90^\circ - \theta) = -\cos\theta$

c. $\sin(90^\circ - \theta) = \sin\theta$

d. $\sin(90^\circ - \theta) = \tan\theta$

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28 [Foundational] • Target: Cofunction Identities and Reciprocals

If $\cos(28^\circ) \approx 0.8829$, what is $\sin(62^\circ)$?

a. 0.8829

b. 0.1171

c. 0.4695

d. 1.1326

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cofunction Identities and Reciprocals

Simplify $\cot(90^\circ - \theta) \cdot \cot\theta$.

a. 1

b. $\tan^2\theta$

c. $\cot^2\theta$

d. 0

30 [Intermediate] • Target: Angles of Elevation and Depression

From a point 50 ft from the base of a building, the angle of elevation to the top is 60°. How tall is the building?

a. $50\sqrt{3}$ ft

b. 100 ft

c. 50 ft

d. $25\sqrt{3}$ ft

31 [Intermediate] • Target: Angles of Elevation and Depression

A lighthouse keeper 120 ft above sea level spots a boat at an angle of depression of 30°. How far is the boat from the base of the lighthouse?

a. $120\sqrt{3}$ ft

b. $60\sqrt{3}$ ft

c. 240 ft

d. 120 ft

32 [Intermediate] • Target: Angles of Elevation and Depression

A $20-$ft ladder leans against a wall reaching a height of 16 ft. What is $\sin\theta$ where θ is the angle the ladder makes with the ground?

a. 4/5

b. 3/5

c. 4/3

d. 3/4

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33 [Intermediate] • Target: Trigonometric ratios

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

34 [Intermediate] • Target: Applications

If $\cos\theta = -\frac{1519}{1681}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1600}{1681}$

b. $-\frac{40}{41}$

c. $\frac{40}{41}$

d. $\frac{441}{410}$

35 [Intermediate] • Target: Solving right triangles

If $\cos\theta = -\frac{3479}{3721}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{661}{610}$

b. $-\frac{60}{61}$

c. $\frac{60}{61}$

d. $\frac{3600}{3721}$

36 [Intermediate] • Target: Angles of elevation and depression

If $\cos\theta = \frac{3479}{3721}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $-\frac{11}{61}$

b. $\frac{171}{610}$

c. $\frac{121}{3721}$

d. $\frac{11}{61}$

37 [Intermediate] • Target: The reciprocal trig ratios

If $\cos\theta = -\frac{119}{169}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{133}{130}$

b. $-\frac{12}{13}$

c. $\frac{12}{13}$

d. $\frac{144}{169}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multi-step surveying problems • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

If $\cos\theta = -\frac{527}{625}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{576}{625}$

b. $\frac{24}{25}$

c. $-\frac{24}{25}$

d. $\frac{53}{50}$

39 [Challenging] • Target: The 30-60-90 and 45-45-90 special triangles • Cross-Lesson: 7.04 Applications: Growth and Decay

If $\cos\theta = -\frac{161}{289}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{225}{289}$

b. $-\frac{15}{17}$

c. $\frac{167}{170}$

d. $\frac{15}{17}$

40 [Challenging] • Target: Trigonometric ratios beyond the first quadrant (preview) • Cross-Lesson: 8.01 Rational Functions and Asymptotes

If $\cos\theta = \frac{1519}{1681}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{9}{41}$

b. $-\frac{9}{41}$

c. $\frac{81}{1681}$

d. $\frac{131}{410}$

41 [Challenging] • Target: The Six Trigonometric Ratios • Cross-Lesson: 8.02 Graphing Rational Functions

If $\cos\theta = -\frac{41}{841}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{239}{290}$

b. $\frac{441}{841}$

c. $\frac{21}{29}$

d. $-\frac{21}{29}$

42 [Challenging] • Target: Special Right Triangles (30-60-90 and 45-45-90) • Cross-Lesson: 8.03 Introduction to Limits

If $\cos\theta = \frac{119}{169}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{25}{169}$

b. $-\frac{5}{13}$

c. $\frac{5}{13}$

d. $\frac{63}{130}$

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43 [Challenging] • Target: Cofunction Identities and Reciprocals • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\cos\theta = \frac{527}{625}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{7}{25}$

b. $\frac{19}{50}$

c. $-\frac{7}{25}$

d. $\frac{49}{625}$

44 [Challenging] • Target: Angles of Elevation and Depression • Cross-Lesson: 9.02 Geometric Sequences and Series

If $\cos\theta = \frac{161}{289}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{8}{17}$

b. $\frac{64}{289}$

c. $\frac{97}{170}$

d. $-\frac{8}{17}$

45 [Challenging] • Target: Trigonometric ratios • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{41}{841}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{229}{290}$

b. $\frac{20}{29}$

c. $-\frac{20}{29}$

d. $\frac{400}{841}$

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